

Reinforcement Learning-Based Robotic Path Planning

Project Overview

This project was conducted as part of my master's Thesis at the Fraunhofer Institute for Casting, Composite and Processing Technology (IGCV) in cooperation with Hochschule Furtwangen University (HFU). The work was embedded within the AutASa (Automatisiertes Abfallsammelfahrzeug) research project, which focuses on developing semi-autonomous robotic systems capable of operating safely and efficiently in dynamic, real-world environments.

The core objective of this project was to design and evaluate a reinforcement learning (RL) based path planning approach for robotic manipulation tasks in environments characterized by dynamic obstacles, uncertainty, and unstructured layouts and conditions where classical deterministic planning methods often fail.

Problem Statement

Traditional robotic path planning techniques (e.g., rule-based or deterministic planners) perform well in static and highly structured environments but struggle in scenarios involving:

- Dynamic and unpredictable obstacles
- Limited prior knowledge of the environment
- Real-time adaptation requirements

Within the context of the AutASa project, a robotic arm mounted on a waste collection vehicle must operate in public and semi-structured environments, interacting safely with static objects and moving entities (e.g., humans). This creates a need for adaptive, learning-based control strategies that can balance goal-reaching performance with safety constraints.

Research Objectives

The main objectives of this project were:

- To develop an RL-based path planning solution for robotic manipulation in dynamic environments.
- To investigate how observation space design influences learning performance and safety.
- To improve collision avoidance and adaptability without sacrificing task accuracy.
- To evaluate system performance using meaningful, application-driven metrics.

Methodology & Technical Approach

Robotic Platform & Simulation

1. Robot: Franka Emika Panda (7-DOF robotic arm)
2. Simulation Environment: PyBullet physics engine
3. Control Space: Cartesian space control
4. Environment Features:
 - Fixed but randomized goal positions
 - Static obstacles
 - Dynamic obstacles with unpredictable motion patterns

PyBullet was selected due to its efficient collision detection, parallel simulation capability, and suitability for iterative reinforcement learning experiments.

Reinforcement Learning Framework

1. Algorithm: Soft Actor-Critic (SAC)
2. RL Library: Stable-Baselines3
3. Key Characteristics:
 - Continuous action space handling
 - Off-policy learning with improved sample efficiency
 - Entropy regularization to encourage exploration and robustness

Observation Space Engineering

A central contribution of this work was the systematic design and optimization of the observation space, progressing through multiple configurations:

1. Distance-Augmented Observation Space
 - End-effector position and velocity
 - Goal position
 - Relative distances to obstacles
2. Goal and Obstacle-Oriented Observation Space
 - Relative positions of obstacles with respect to the end-effector
 - Goal-directed features
3. Enhanced Observation Space (Final Design)
 - Line-of-sight indicators between end-effector and goal
 - Obstacle proximity and motion cues
 - Context-aware features enabling adaptive path deviation

This iterative refinement significantly improved the agent's ability to reason about spatial constraints and dynamic interactions.

Hyperparameter Optimization

Key SAC hyperparameters were systematically tuned, including:

- Learning rate
- Batch size
- Network architecture (MLP, LSTM, CNN variants)

This ensured stable training and consistent policy convergence across scenarios.

Evaluation Metrics

System performance was evaluated over 10,000 episodes using three primary metrics:

- Safe Rate: Frequency of collision-free trajectories.
- Accuracy: Ability to reach the target without collision.
- Cumulative Reward: Balance between task efficiency and safety behavior.

These metrics were selected to reflect practical deployment requirements, rather than purely algorithmic performance.

Results & Key Outcomes

1. Achieved significant improvements in safety and accuracy through observation space optimization.
2. Demonstrated robust behavior in scenarios with dynamic obstacles of varying size, shape, and motion.
3. Enabled real-time path adaptation and collision-aware decision-making.
4. Validated the effectiveness of RL-based solutions for applied industry-driven robotic applications.

The final observation space configuration consistently outperformed baseline designs across all evaluated scenarios.

Key Contributions

1. Designed an RL-based robotic path planning framework tailored for dynamic environments.
2. Demonstrated the critical role of observation space engineering in safety and adaptability.
3. Delivered an industry-relevant solution aligned with Fraunhofer IGCV research objectives.
4. Established a foundation for future work in sim-to-real transfer and joint-space control.

Future Outlook

Potential extensions of this work include:

- Transition from Cartesian to joint-space control for enhanced dexterity.
- Further refinement of reward structures to balance safety and efficiency.
- Domain randomization and sim-to-real transfer studies.
- Integration with real robotic hardware platforms.

Tools & Technologies

1. Python
2. PyBullet
3. Stable-Baselines3
4. Soft Actor-Critic (SAC)
5. Franka Emika Panda
6. Reinforcement Learning & Robotics Simulation